

Maag

Modular Agentic Autonomous Greenhouse

27 steps · tools · wiring

Gantry assembly instructions, Rev E. The X/Z gantry and the myCobot 280 arm for the 1397 × 559 × 1829 cabinet.

Rendered from the same models the steps are drawn from: Elephant Robotics' arm and gripper meshes, OpenBuilds' C-Beam parts.

Gantry · parts

Count everything first. Quantities cover the complete gantry.



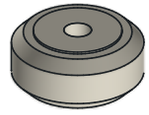
C-Beam 1500, silver

2×



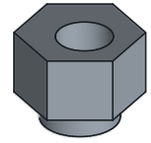
XL gantry plate 125 × 125 × 6

2×



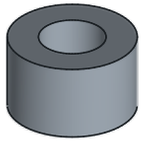
Xtreme Solid V wheel

8×



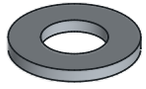
Eccentric spacer, 10 mm hex

4×



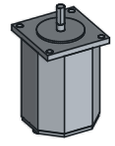
Aluminium spacer 6 mm

6×



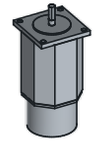
Precision shim 10 × 5 × 1

8×



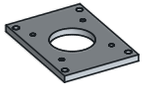
NEMA 23 · X axis

1×



NEMA 23 + 24 V brake · Z

1×



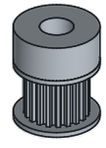
NEMA 23 motor plate

2×



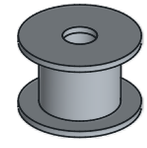
C-Beam End Mount

1×



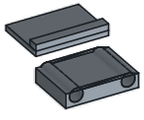
GT2 pulley 20T, 6.35 bore

1×



Idler, 9 mm belt

1×

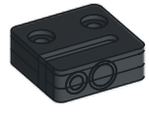


Belt clamp block

2×

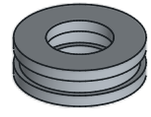
Lead screw Tr8×8, 1500

1×



Anti-backlash nut block

1×



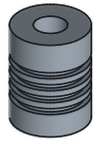
Thrust bearing 8 × 16 × 5

2×



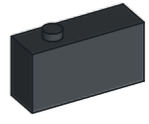
Lock collar 8 mm, 14 × 7

2×



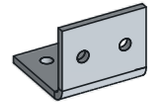
Coupler 6.35 → 8

1×



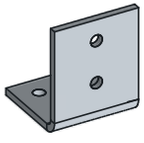
Limit switch, NC

4×



Rail bracket, top · 5052

2×



Rail bracket, bottom

2×



Arm plate Rev B, 160 × 6.35

1×

Also: GT2 belt 9 mm (2.8 m for the loop), the wiring on the wiring pages, fuse block, E-stop and the safety relay.

In the BOM, not drawn. Rev E changes: shims, the End Mount, the spacer count and the bent rail brackets.

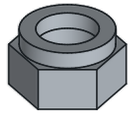
Gantry · fasteners

Every screw in the build. Lay them out in labelled pots first.



M5 × 30 low profile · wheels

8×



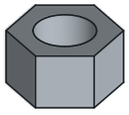
M5 nyloc · wheels, idler

9×



M5 × 25 low profile · block, clamps, mount

10×



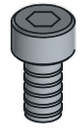
M5 hex nut · nut block

2×



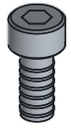
M5 × 10 low profile · mast to X carriage

6×



M5 × 10 socket cap · motors, X motor plate

12×

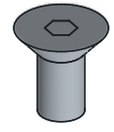


M5 × 12 socket cap · Z foot plate

4×

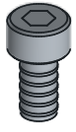
M5 × 40 socket cap · idler axle

1×



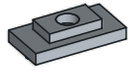
M5 × 10 flat head · arm plate

4×



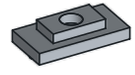
M5 × 10 socket cap · rail brackets

16×



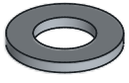
M5 T-nut, 20 series · C-Beam slots

14×



M5 T-nut, slot 8 · 4040 frame

8×



M5 washer · idler

2×



M5 × 6 grub · pulley

2×



M4 × 6 grub · coupler

4×



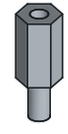
M3 × 35 socket cap · arm G base

4×



M3 × 10 + T-nut · limit switches

8×



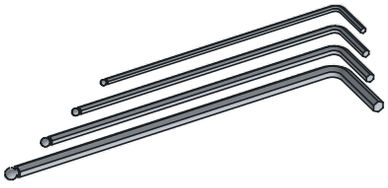
M3 standoff, nut, M3 × 6 · controller

4×

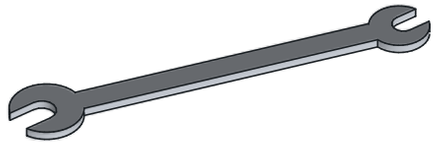
OpenBuilds low profile heads are only 1.5 mm tall: that is what lets them sit between a plate and the beam (2.5 mm gap) and inside the plate's 1.6 mm counterbores. Do not substitute DIN 7984 heads (2.8 to 3.5).

Tools

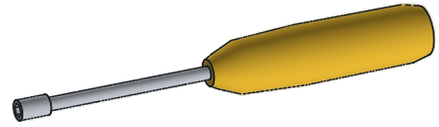
Everything the 27 steps and the wiring need. Metric throughout; nothing is inch.



Hex keys, ball end
2 · 2.5 · 3 · 4 mm, long series
steps: 2: 14 · 2.5: 7, 13, 15, 16, 19 (a long one)
3: 2, 3, 10, 11, 13, 18, 24 · 4: 6, 8, 9, 14, 23



Thin open spanner
8 × 10 mm, under 3.5 mm thick
steps: 8 mm: M5 nylocs and nuts, 2, 3, 9, 11
10 mm: the eccentric spacers, 3 and 4



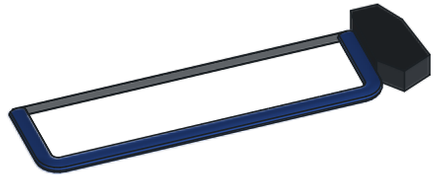
Nut driver
5.5 mm
steps: M3 nuts: limit switches and
controller standoffs, 15 and 16



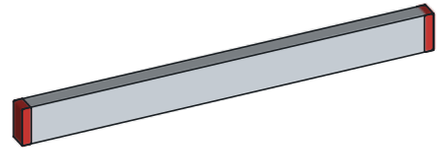
Drill and 5.2 mm bit
HSS jobber, plus a centre punch
steps: the idler axle hole
through the rail's web, 9



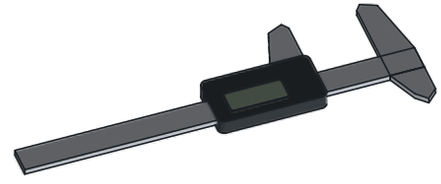
M5 tap and tap wrench
M5 × 0.8, with cutting oil
steps: C-Beam end holes, unless bought
tapped: 8, 13, 14; plain plate holes, 6



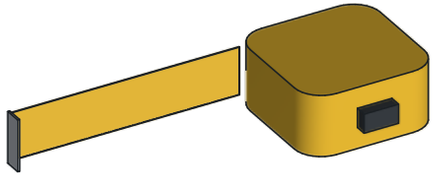
Saw
hacksaw and mitre box, or a mitre saw
steps: the X rail, 1500 cut to 1397, 23.
Non-ferrous blade; deburr the cut



Spirit level
600 mm, with a plumb vial
steps: the cabinet, 21; the X rail, 23;
the mast, 24



Calipers
150 mm, digital
steps: pulley hub 22 to 32 mm, 7; the 2 mm
coupler gap, 14; checking stock



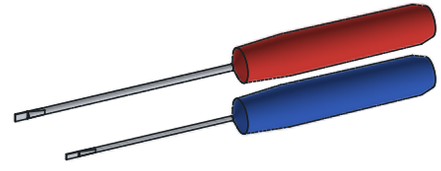
Tape measure
5 m
steps: rail heights and the cut, 21 to 24



Multimeter
continuity, ohms, DC volts
steps: motor coil pairs, limit loops, the
brake and both 24 V rails: wiring, 14, 17



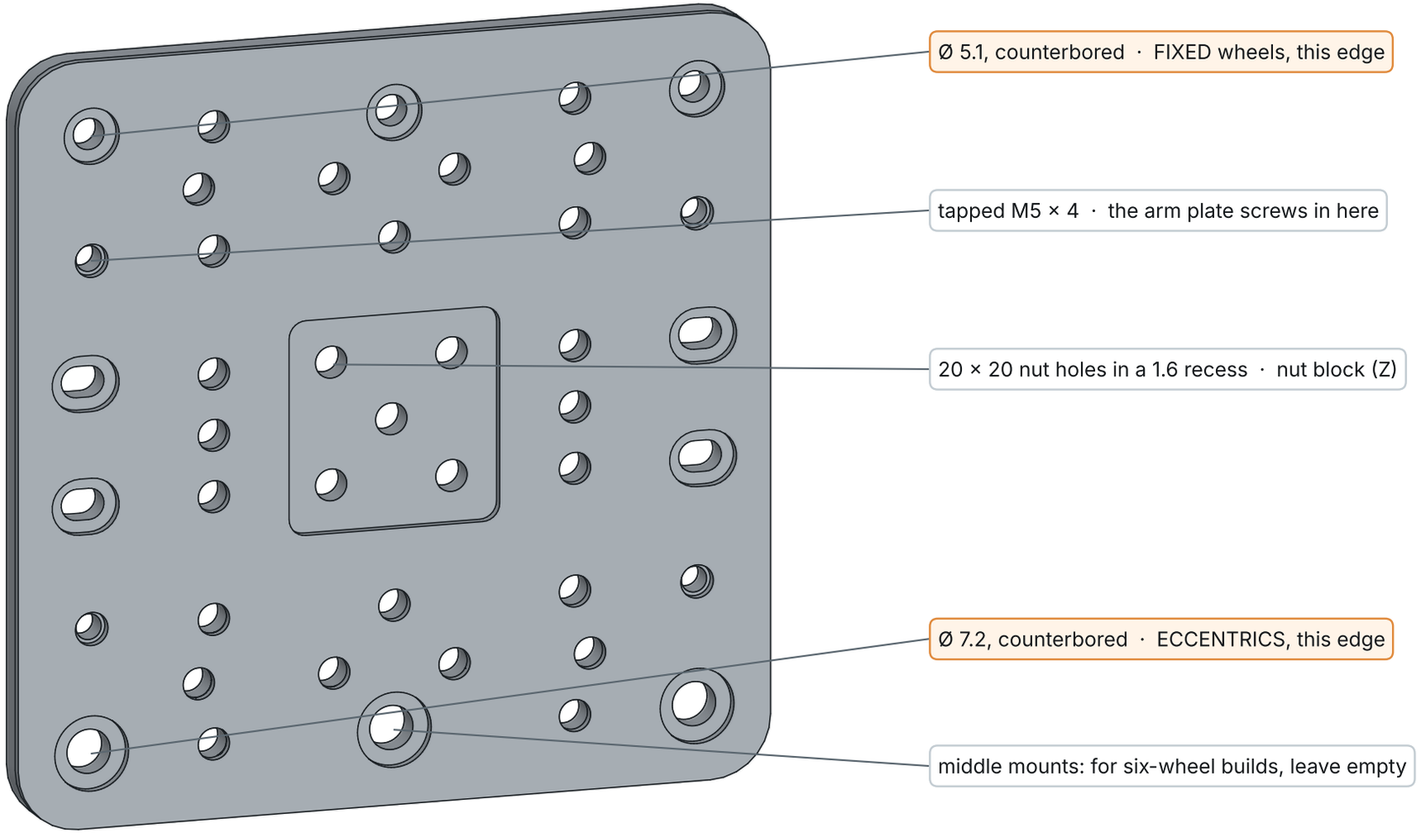
Ferrule crimper and stripper
0.25 to 2.5 mm², square crimp
steps: every screw terminal: wiring pages



Terminal screwdrivers
flat 2.5 and 3.5 mm
steps: relay, fuse block and controller
terminals: wiring pages

1 Know your plate

Fixed wheels on one edge, eccentrics on the other. Use the outer two of each row.

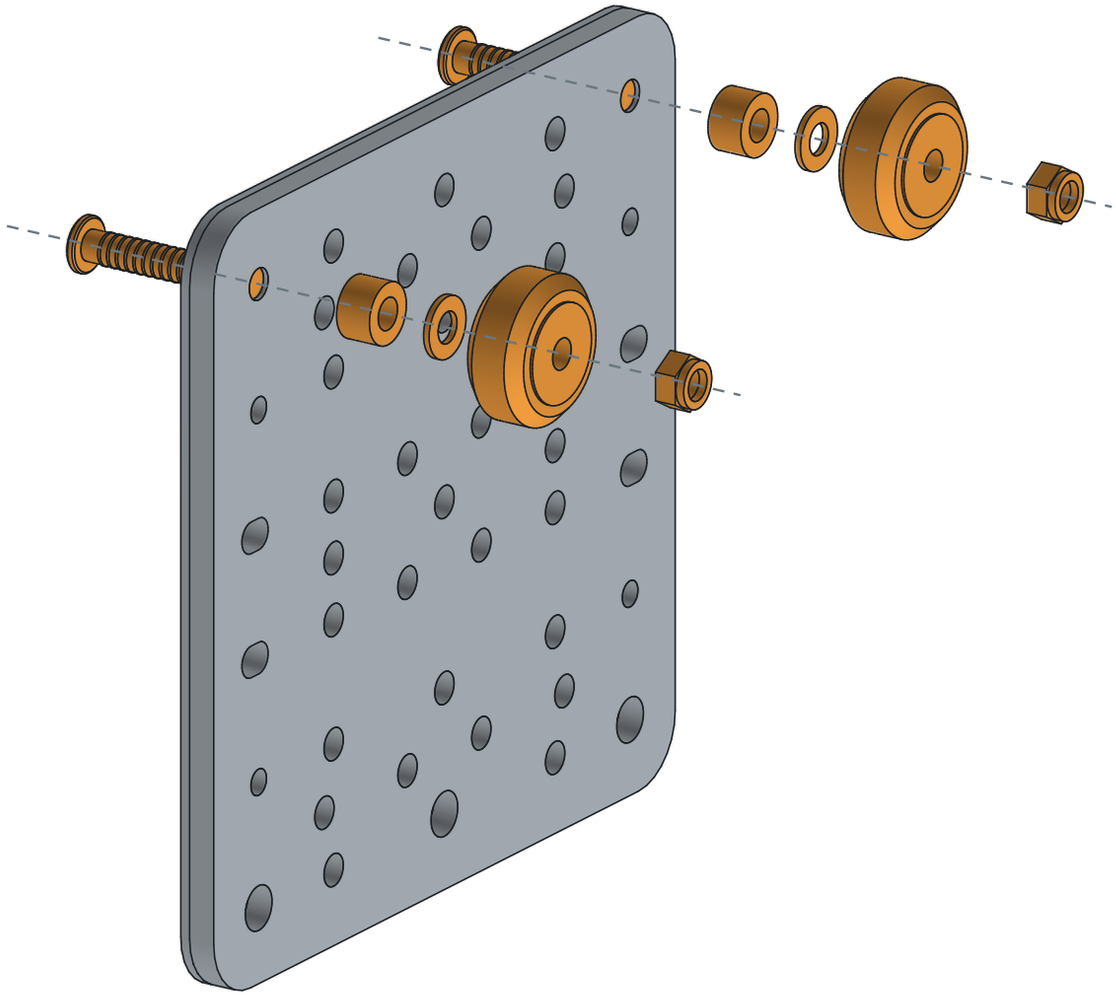


The wheels straddle the beam's 80 mm, 100.6 mm apart. Both FIXED wheels go on one edge and both eccentrics on the other: the plate is drilled that way, and it is what lets two eccentrics squeeze the beam between the rows.

On the X carriage the fixed edge goes UP, so the fixed wheels carry the hanging mast.

2 Fixed wheels · the stack

Screw, plate, spacer, shim, wheel, nyloc. The two outer 5.1 holes.



- 2× M5 × 30 low profile
- 2× Aluminium spacer 6 mm
- 2× Precision shim 10 × 5 × 1
- 2× Xtreme Solid V wheel
- 2× M5 nyloc

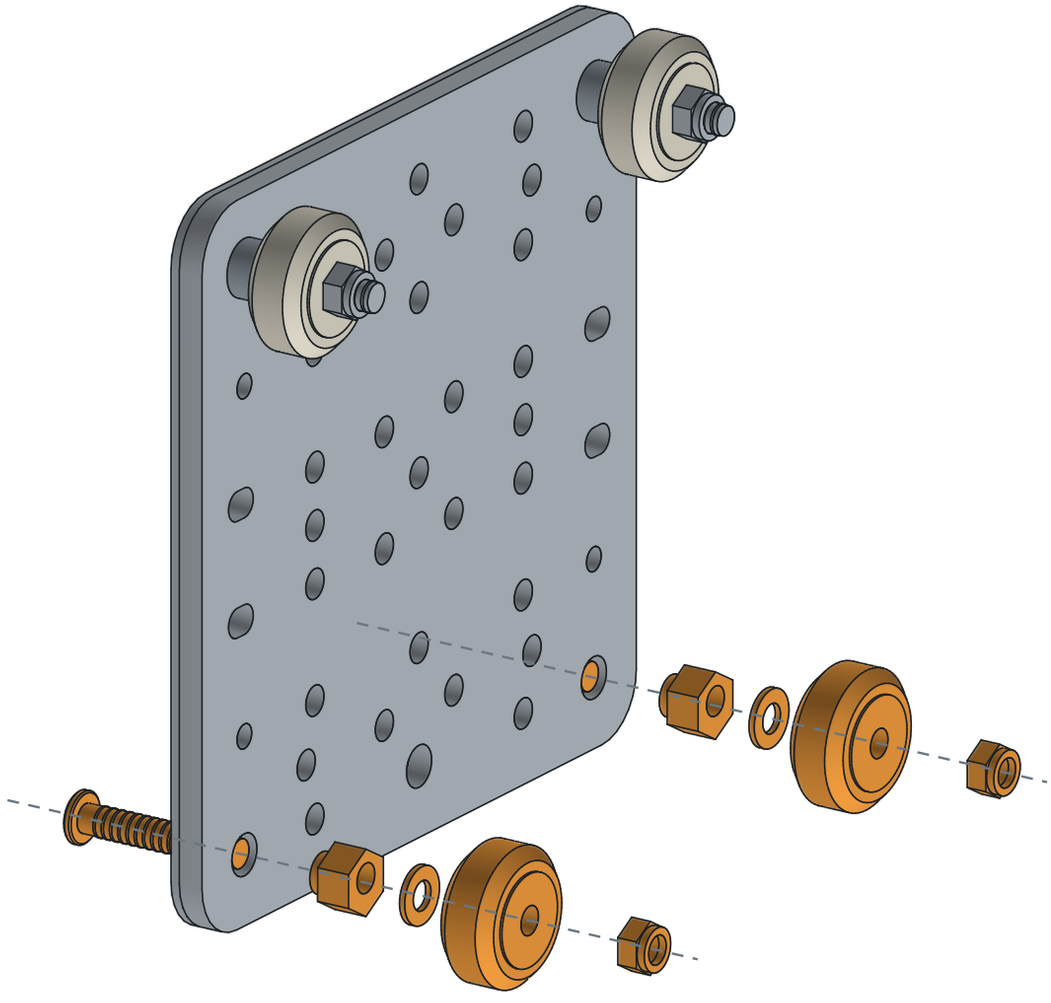
3 mm hex · 8 mm

Heads go in the counterbores, on the plate's outer face. The shim keeps the bearing's outer race off the spacer.

Snug the nylocs, do not torque them: you re-check every one after preload.

3 Eccentric wheels

The opposite edge. The hex sits between the plate and the shim.

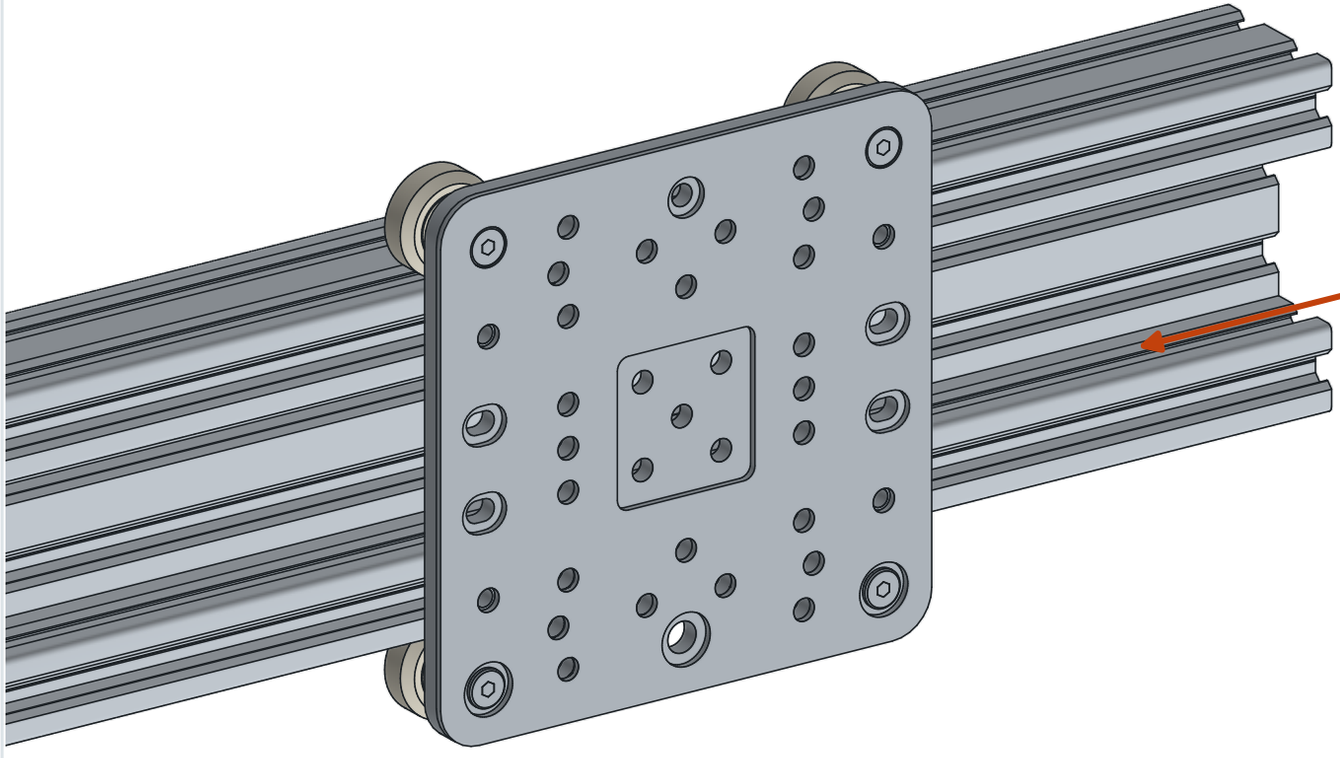


- 2× Eccentric spacer, 10 mm hex
- 2× Precision shim 10 × 5 × 1
- 2× Xtreme Solid V wheel
- 2× M5 × 30 low profile
- 2× M5 nyloc

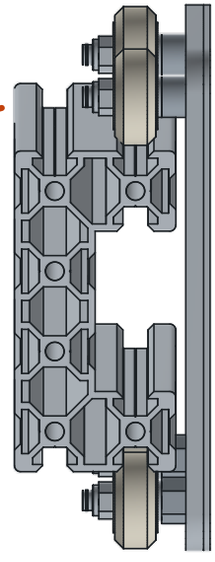
The spacer's 2.5 mm boss drops into the 7.2 mm hole; its hex is the spacer. Set both at their LOOSEST (bore furthest from the beam) before fitting, or the carriage will not go on. A thin 10 mm spanner turns them later.

4 Onto the rail, then set the preload

Slide on from the end. Then the one adjustment that decides the machine.



end view: wheels ride the top and bottom grooves



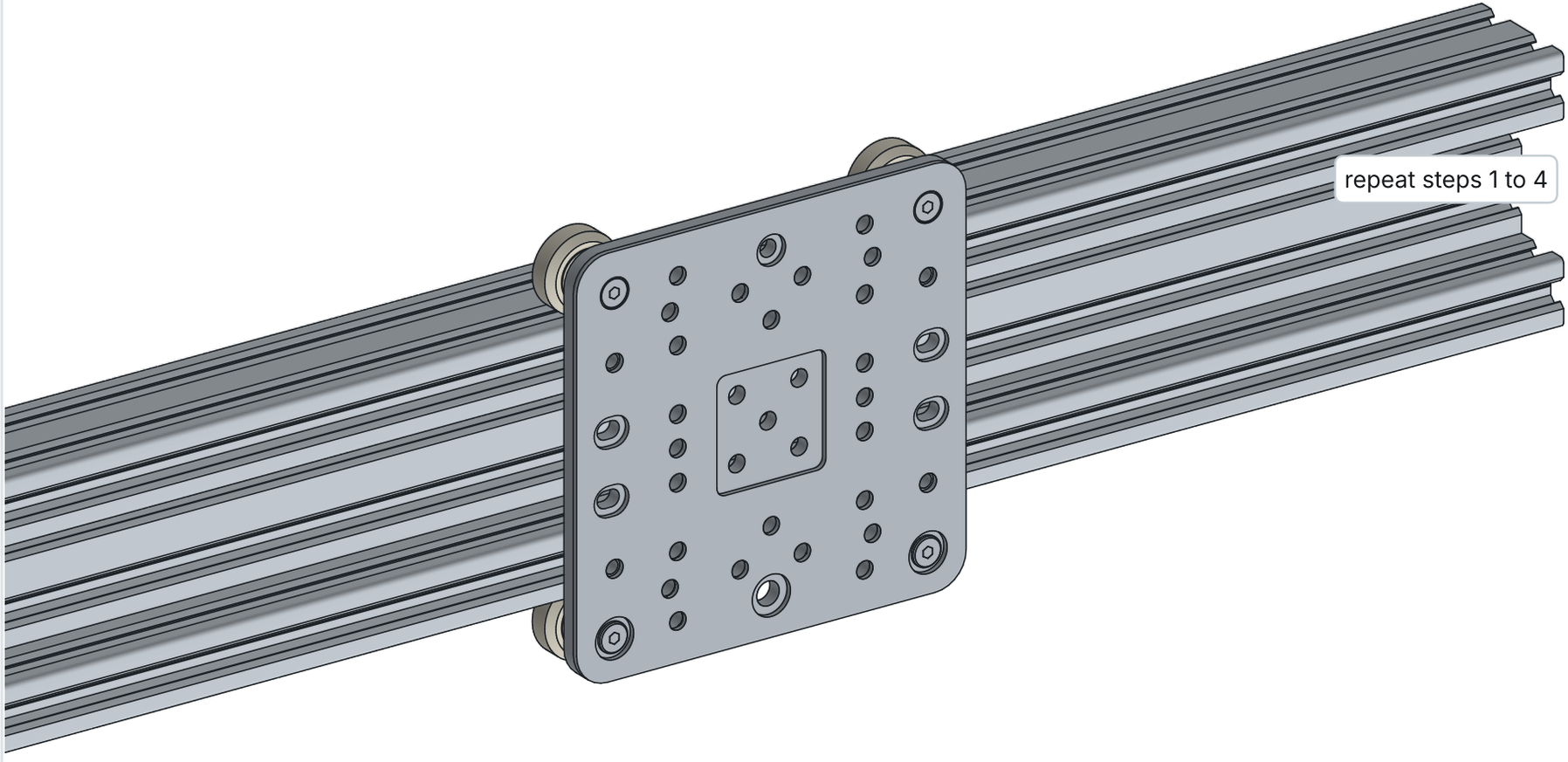
LOOSE	wheel spins free, carriage rocks
RIGHT	turns with light finger drag
TIGHT	hard to turn, wears flats in the wheel

thin 10 mm spanner · 8 mm

Turn each eccentric until its wheel just grips the groove, then back off until it turns with light finger drag. Tighten the nyloc and re-check. Run the carriage the full length by hand: even resistance, no rock, no tight spot. Too loose is slop, and slop defeats hand-eye calibration: the camera says where the leaf is and the arm arrives somewhere else.

5 Build the second carriage

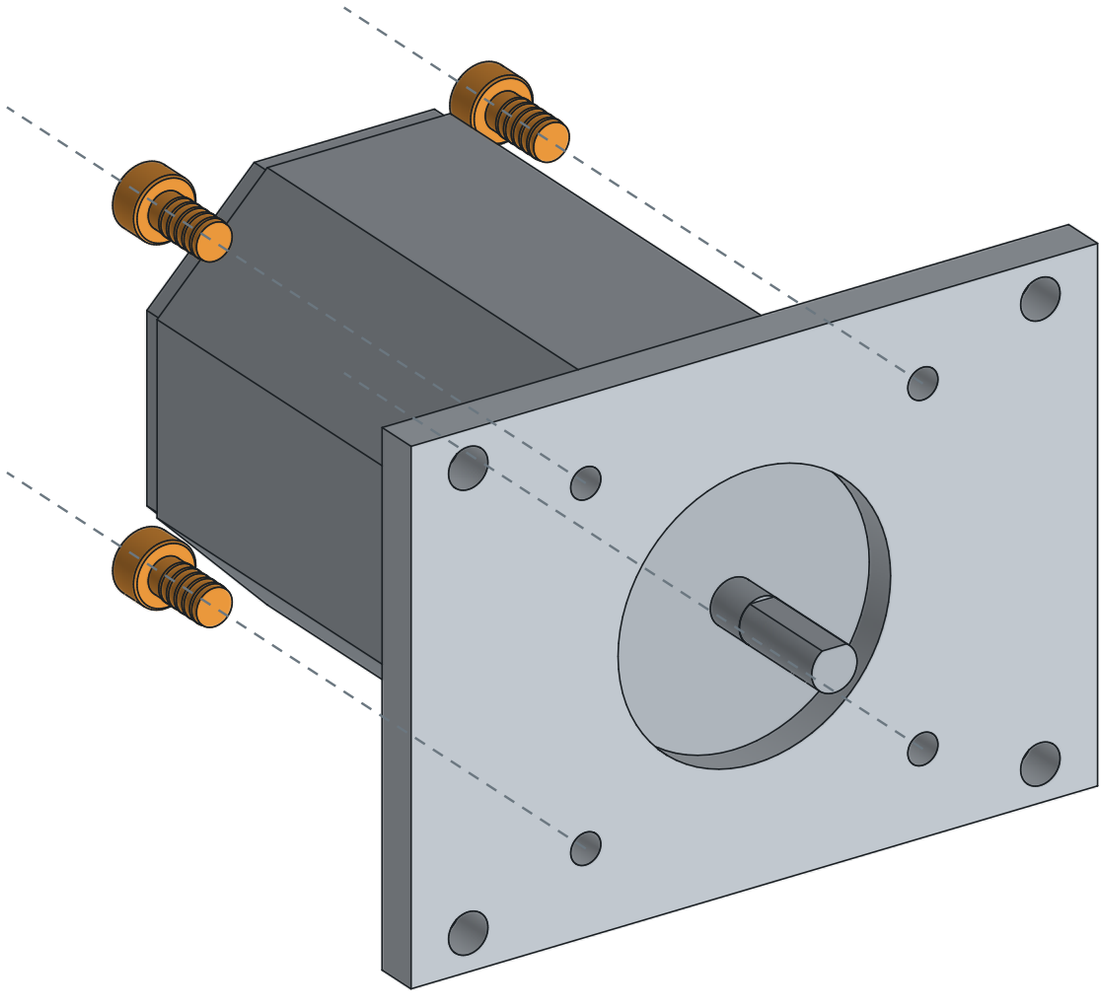
Identical. It rides the 1500 mm Z mast.



Build and preload it on the mast now, while the method is fresh, then slide it off again: at step 11 the nut block goes on its back, and from then on it only goes on from the mast's end.

6 X motor onto its plate

Four M5 × 10, from the MOTOR side, into the plate.



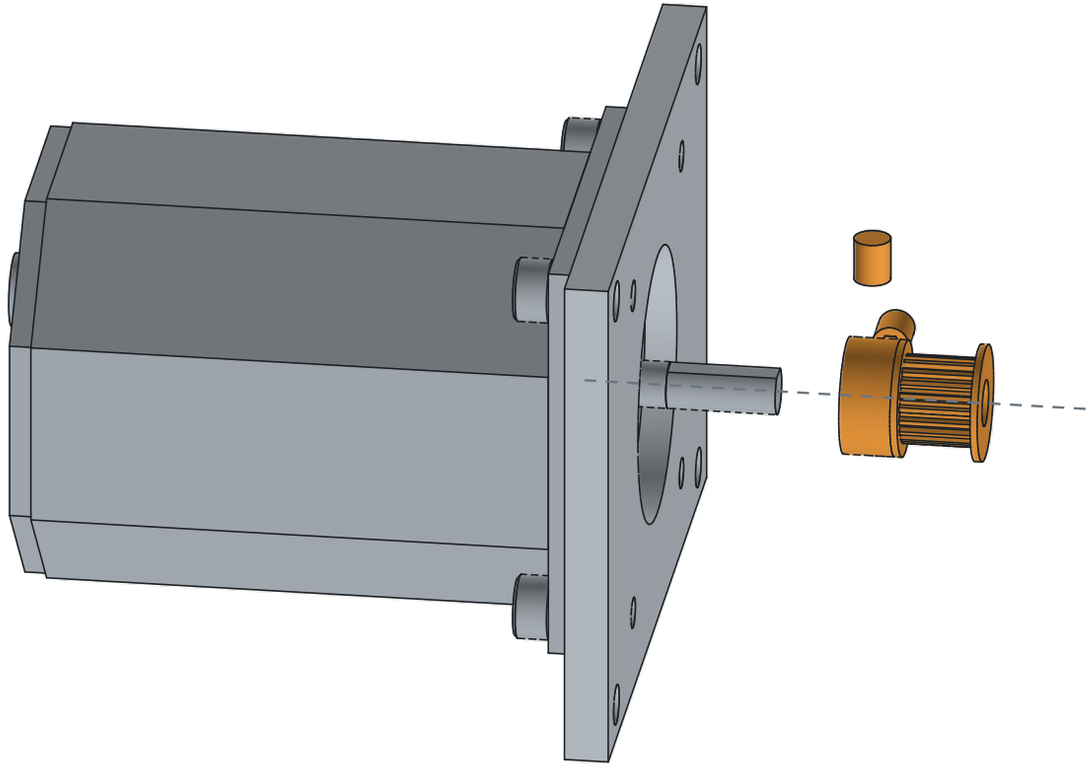
4× M5 × 10 socket cap

4 mm hex

NEMA 23 flanges are drilled through, so the heads seat on the BACK of the motor flange, in the pockets beside the body, and the thread takes in the plate. If the plate's four motor holes are plain, tap them M5 first. The boss locates in the 38.1 bore: if it will not seat, something is between the faces.

7 Pulley onto the shaft

Now, while you can reach the grub screws. After step 8 you cannot.



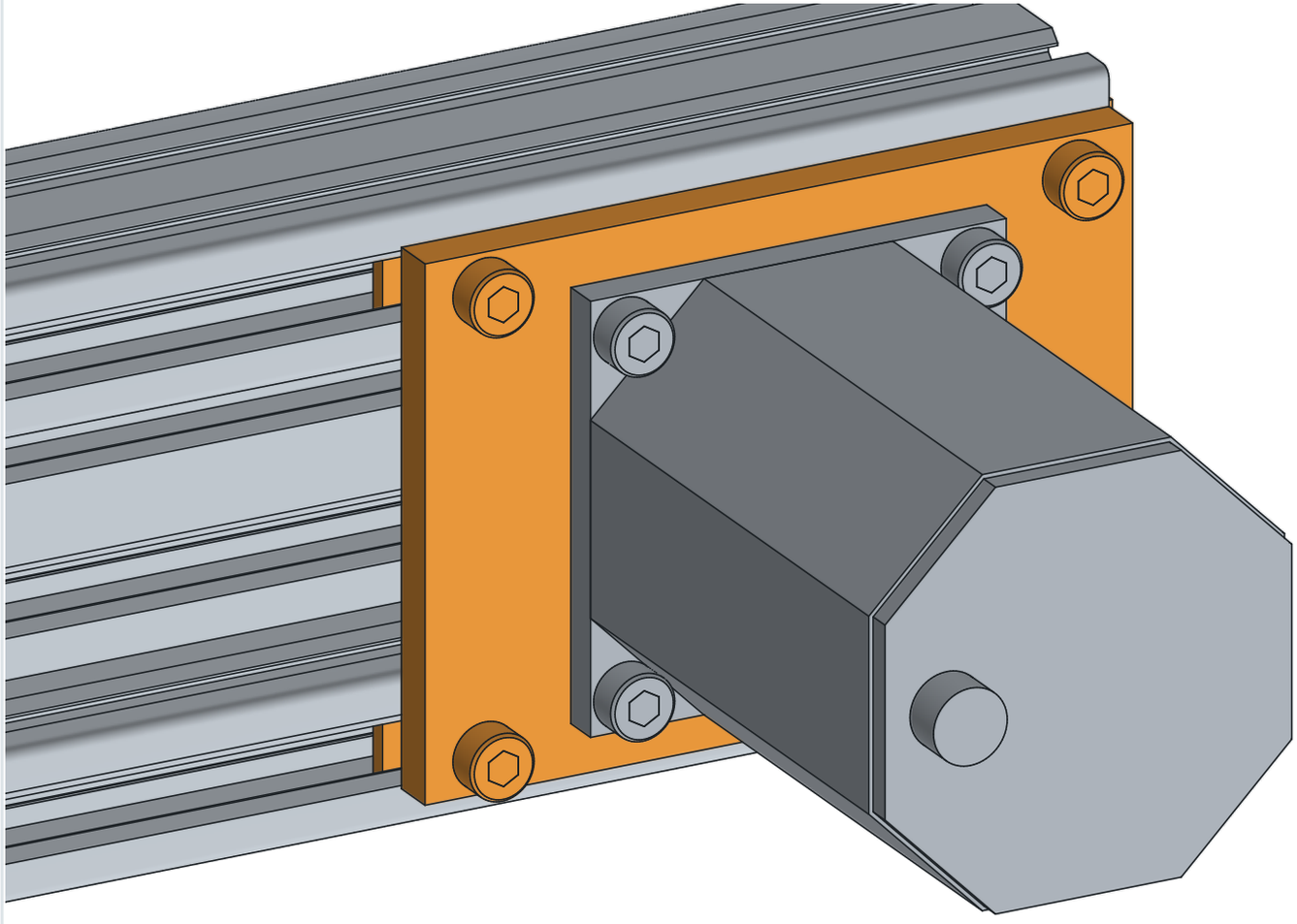
- 1× GT2 pulley 20T, 6.35 bore
- 2× M5 × 6 grub screw

2.5 mm hex

Boss toward the motor, flanges outboard: the toothed hub must finish 22 to 32 mm from the plate's back face, which is where the belt runs inside the rail. Seat the FIRST grub on the shaft flat. A grub on the round creeps under load and loses steps exactly like a current problem would.

8 Motor onto the rail's left end

Four M5 × 10 into T-nuts in the front-lip grooves. Pulley into the channel.

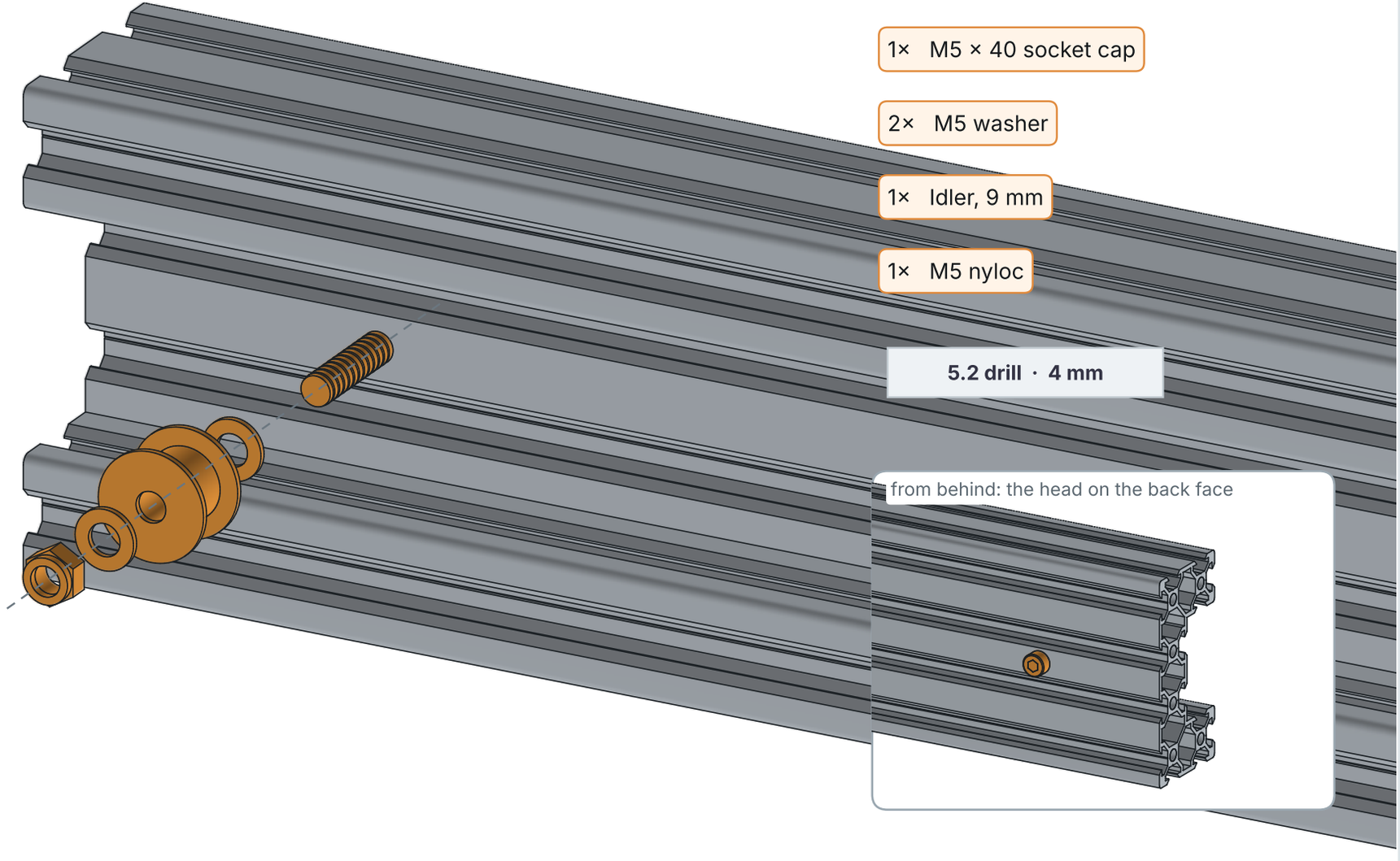


- 4× M5 × 10 socket cap
- 4× M5 T-nut

Flush with the rail's end. The plate sits on the two front lips and its four screws drop into the lip grooves, 60 mm apart. The motor never moves: the belt does the travelling, and the carriage stops 5 mm short of this plate.

9 Idler at the far end

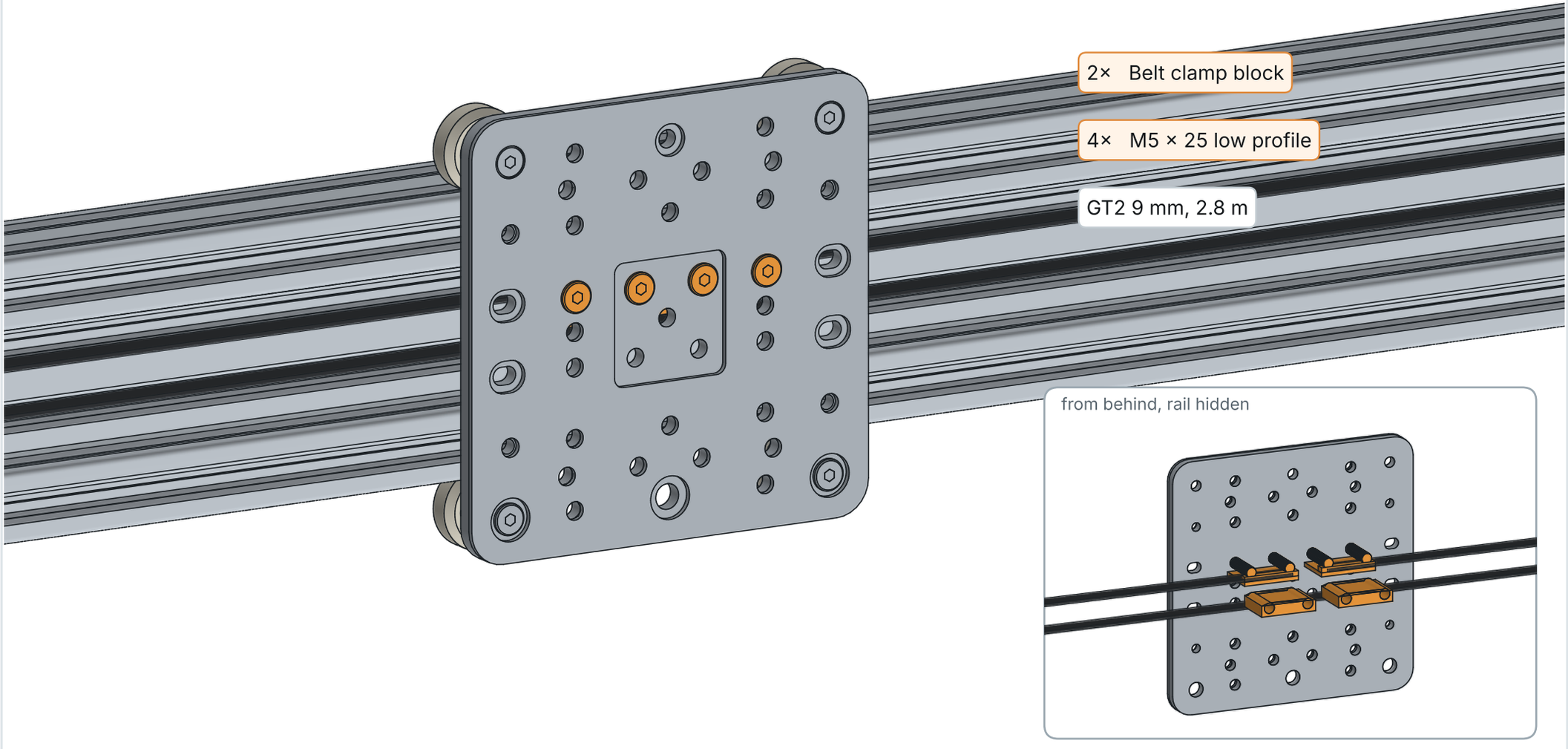
Drill 5.2 through the web, 50 from the end. M5 × 40 from the back.



Head on the back face, then inside the channel: washer, idler, washer, nyloc. Centre the hole on the web's mid-height so the idler lines up with the pulley at the other end. The nyloc snug, not tight: the idler has to spin freely.

10 Belt and clamps

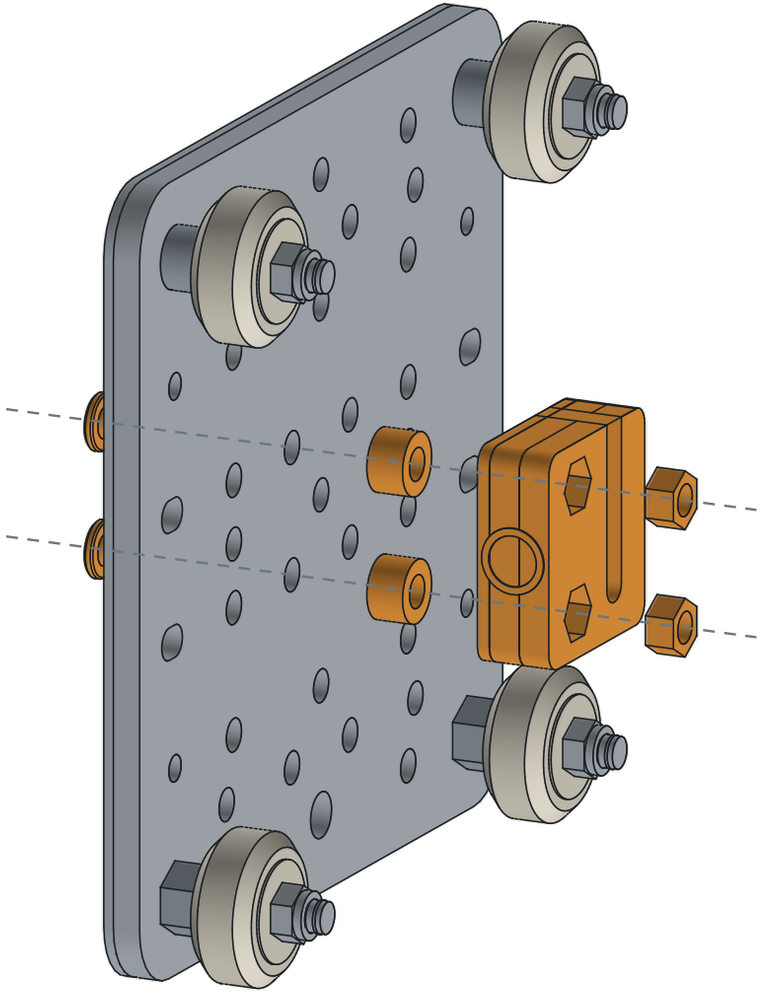
Teeth in, width along the pulley axis. Both ends clamp to the carriage.



Each clamp takes two screws through holes the plate already has: 10 and 30 either side of the centre, on the row 10 above it. Clamp one end. Run the belt left inside the channel, round the pulley, back along the bottom of the channel, round the idler, and into the second clamp. Pull it through to tension and lock: plucked, a low clear note. Over-tension bends the motor shaft.

11 Nut block onto the Z carriage

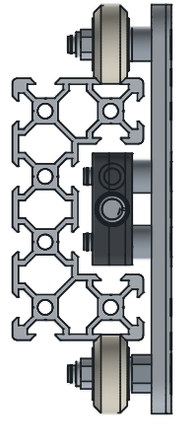
Carriage OFF the mast. The block ends up inside the mast's channel, around the screw.



- 2× M5 × 25 low profile
- 2× Aluminium spacer 6 mm
- 1× Anti-backlash nut block
- 2× M5 hex nut, in the block's pockets

3 mm hex · 8 mm

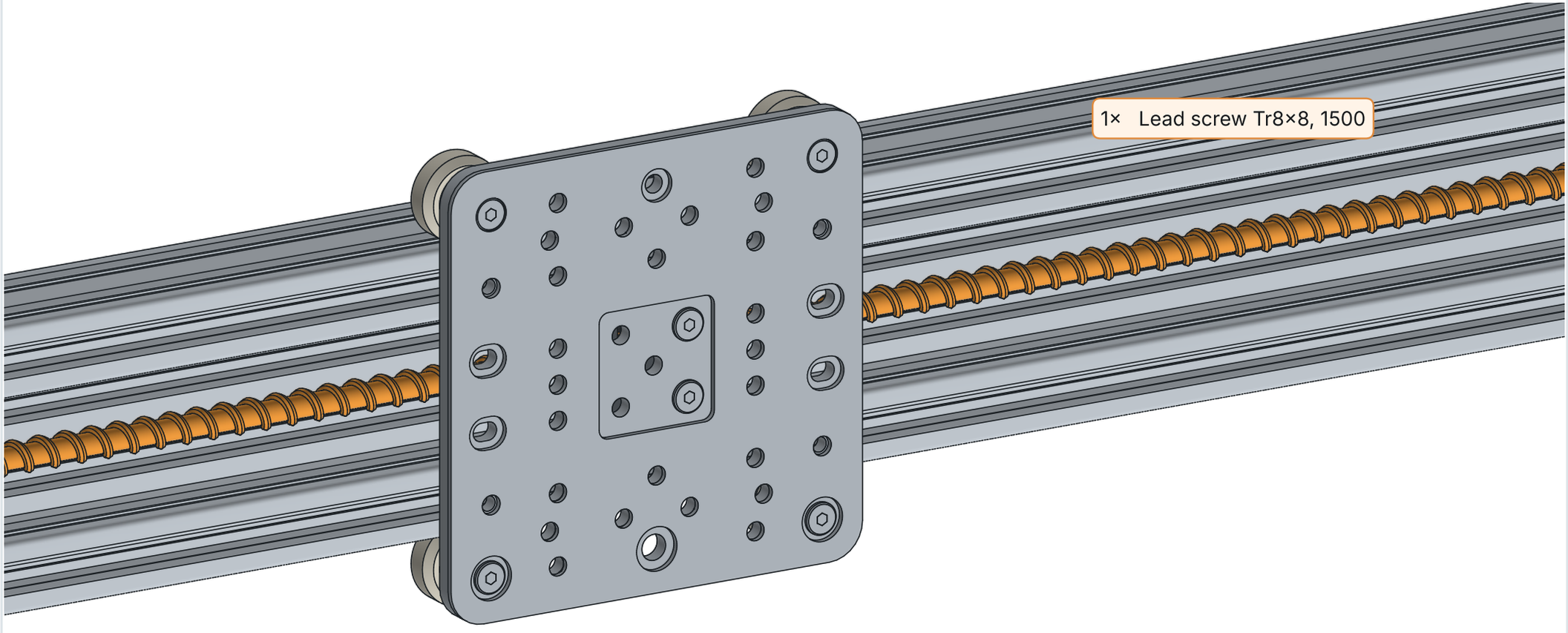
end view down the mast: block in the channel



Screws from the plate's outer face, through the two nut holes nearest one edge of the recess (20 apart, across the mast), then a spacer each, the block, and a nut in each pocket. The bore runs ALONG the mast, 9.5 mm behind its mouth. Then slide the carriage back on from the mast's end and re-check the preload. Leave the block's tension screw for step 12.

12 Lead screw through the nut

Thread it in by hand, through the carriage, along the channel.

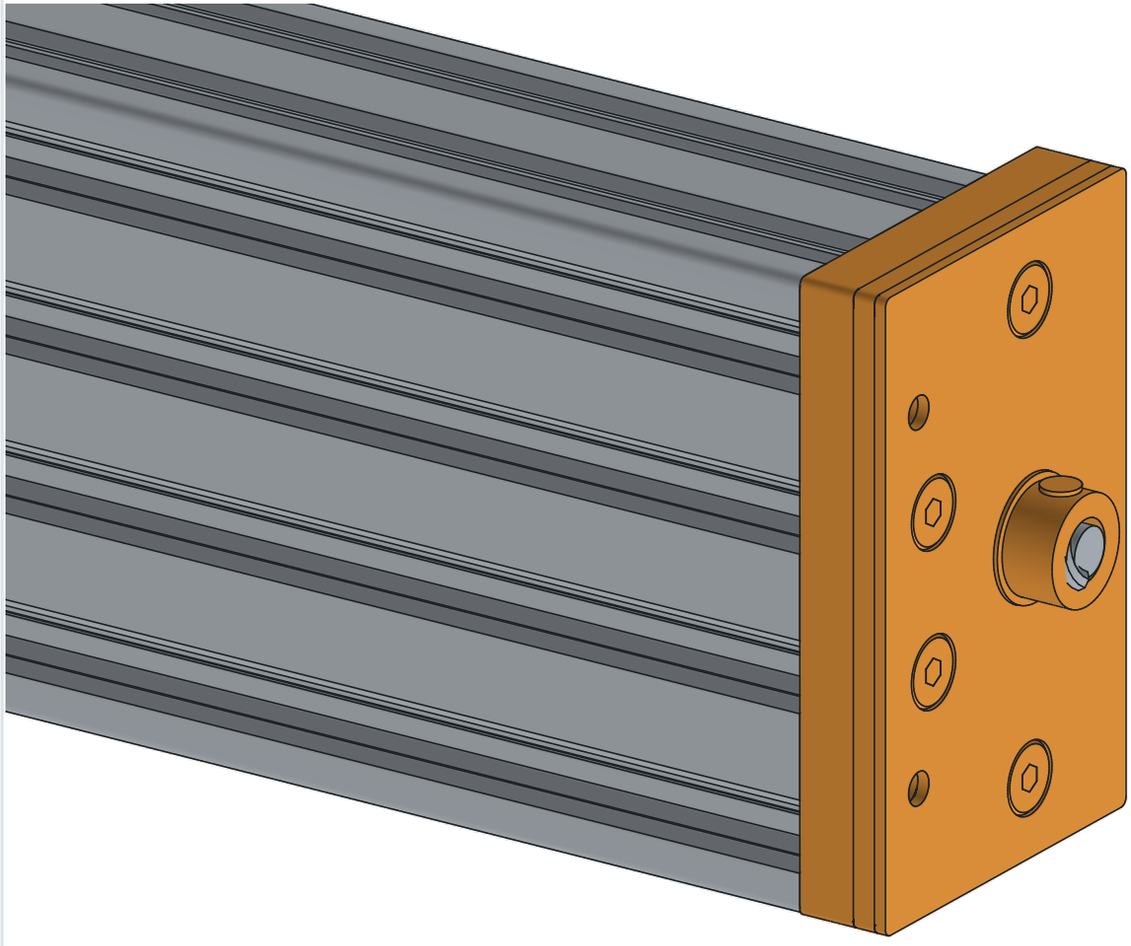


It runs in the channel, 9.5 mm behind the mouth. Turn it through the nut by hand until it runs the full length with even drag, then snug the block's tension screw until the play is gone and the drag is still light.

A tight spot now is a bent screw or a block out of line, both cheaper to find before the motor is on.

13 Top end · the screw hangs here

C-Beam End Mount on the mast's end, a thrust bearing in its recess, one underneath.



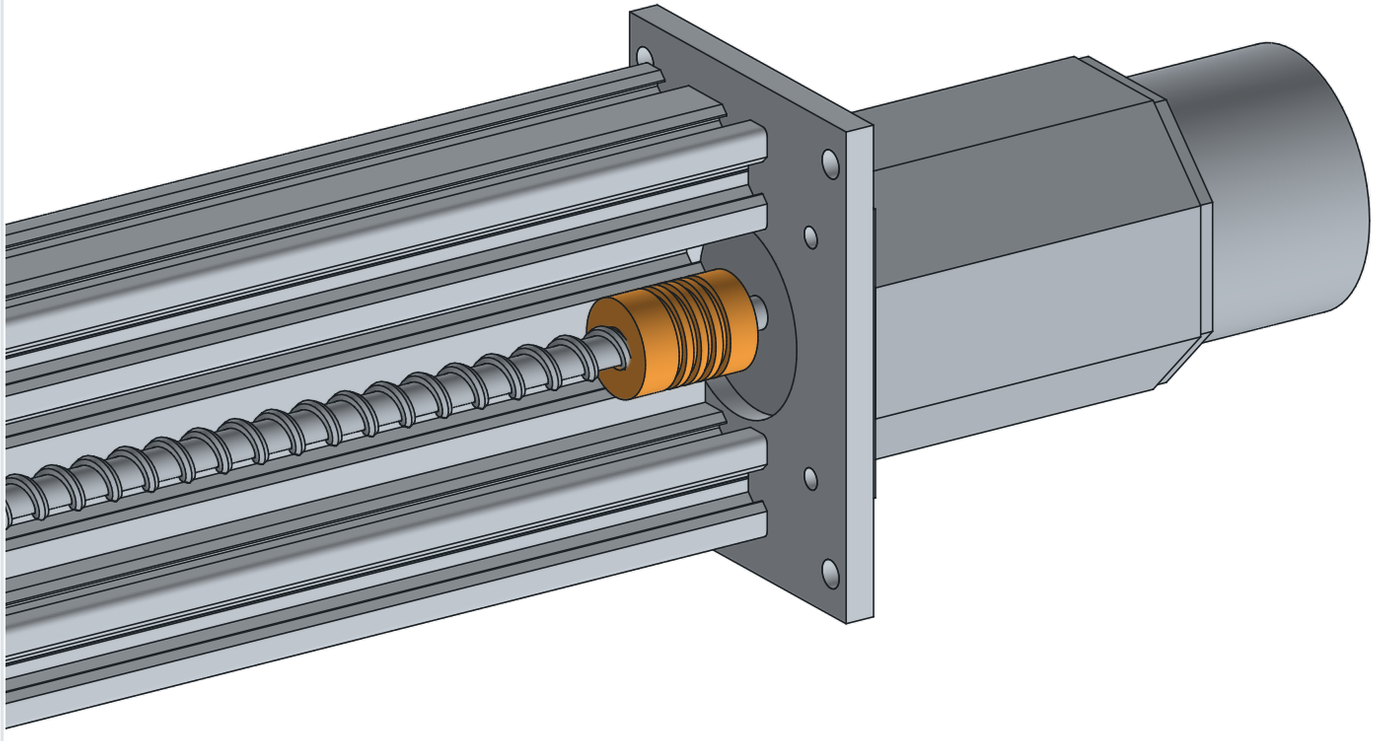
- 1× C-Beam End Mount 80 × 50 × 12
- 4× M5 × 25 low profile, into the end taps
- 2× Thrust bearing 8 × 16 × 5
- 2× Lock collar 8 mm, 14 × 7

3 mm hex · 2.5 mm hex

The mount overhangs the mast's mouth by 10 mm, toward the carriage. One bearing sits in the 16.1 recess on top with a collar above it; the other goes under the mount with a collar below. Tighten the collars against the bearings with the screw pulled up: then it hangs from here and the coupler at the foot only ever carries torque.

14 Foot · braked motor and coupler

Motor plate across the end, motor outside, coupler inside.



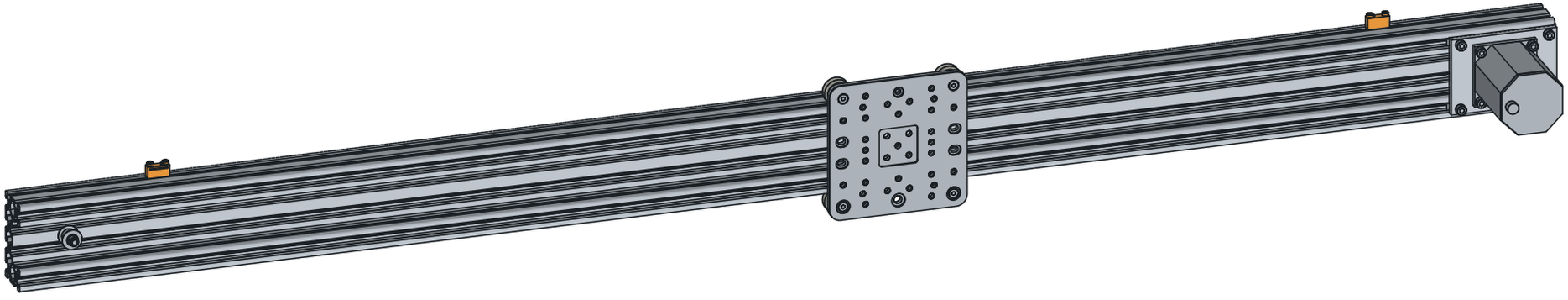
- 1× Coupler 6.35 → 8
- 4× M4 × 6 grub
- 4× M5 × 12 into the end taps

2 mm hex

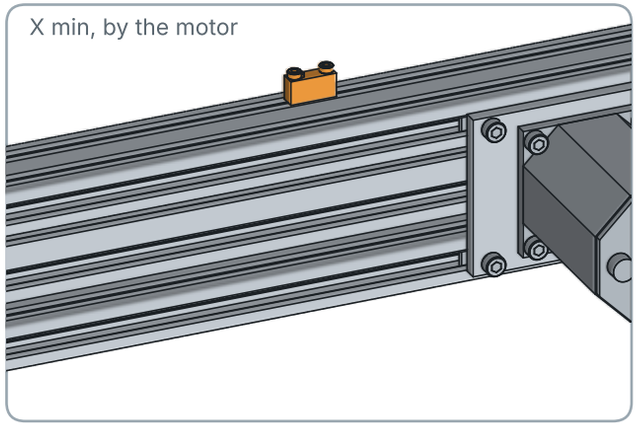
Leave 2 mm between the shaft ends inside the coupler, so it can flex. Test the brake BEFORE coupling: 24 V releases it, no power grips it. If it does not grip you find out when the power blinks and the arm lands on a tray.

15 Limit switches · all four

X min and X max on the rail; Z min and Z max the same way on the mast.

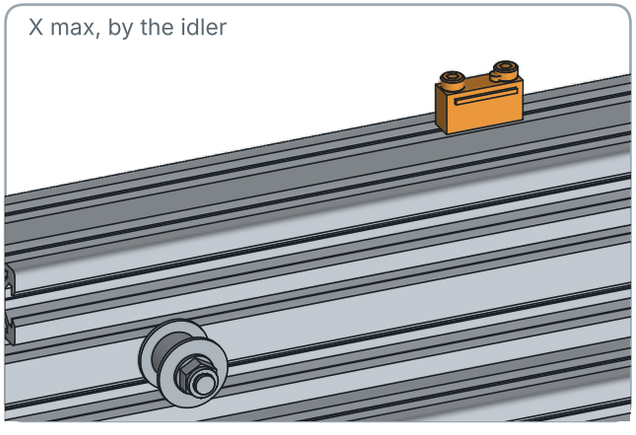


X min, by the motor



8× M3 × 10 + T-nut

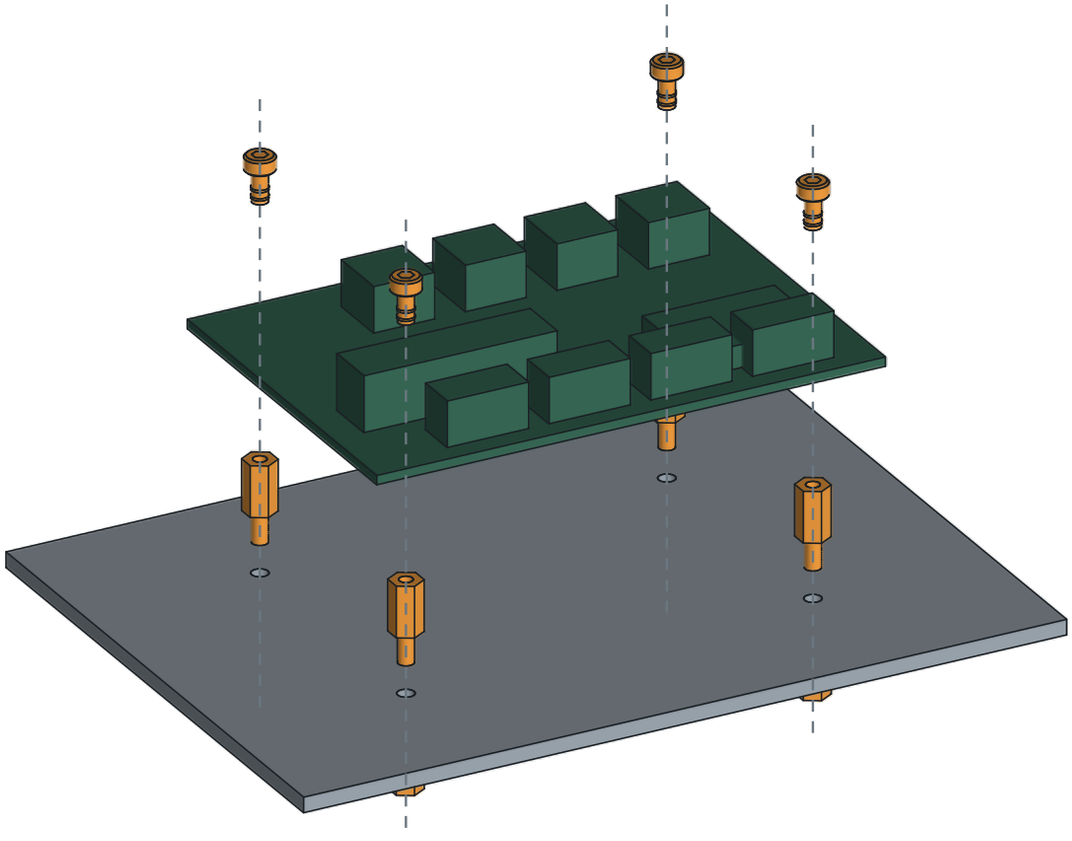
X max, by the idler



In the rail's back top groove: the front one is where the wheels run. Each trips 5 to 10 mm BEFORE the hard stop. Wire them NORMALLY CLOSED, in series per axis: a cut wire then reads as a trip instead of silently disabling the limit.

16 Controller on standoffs

Panel, standoffs, board, screws. Keep the board's ground off the frame.



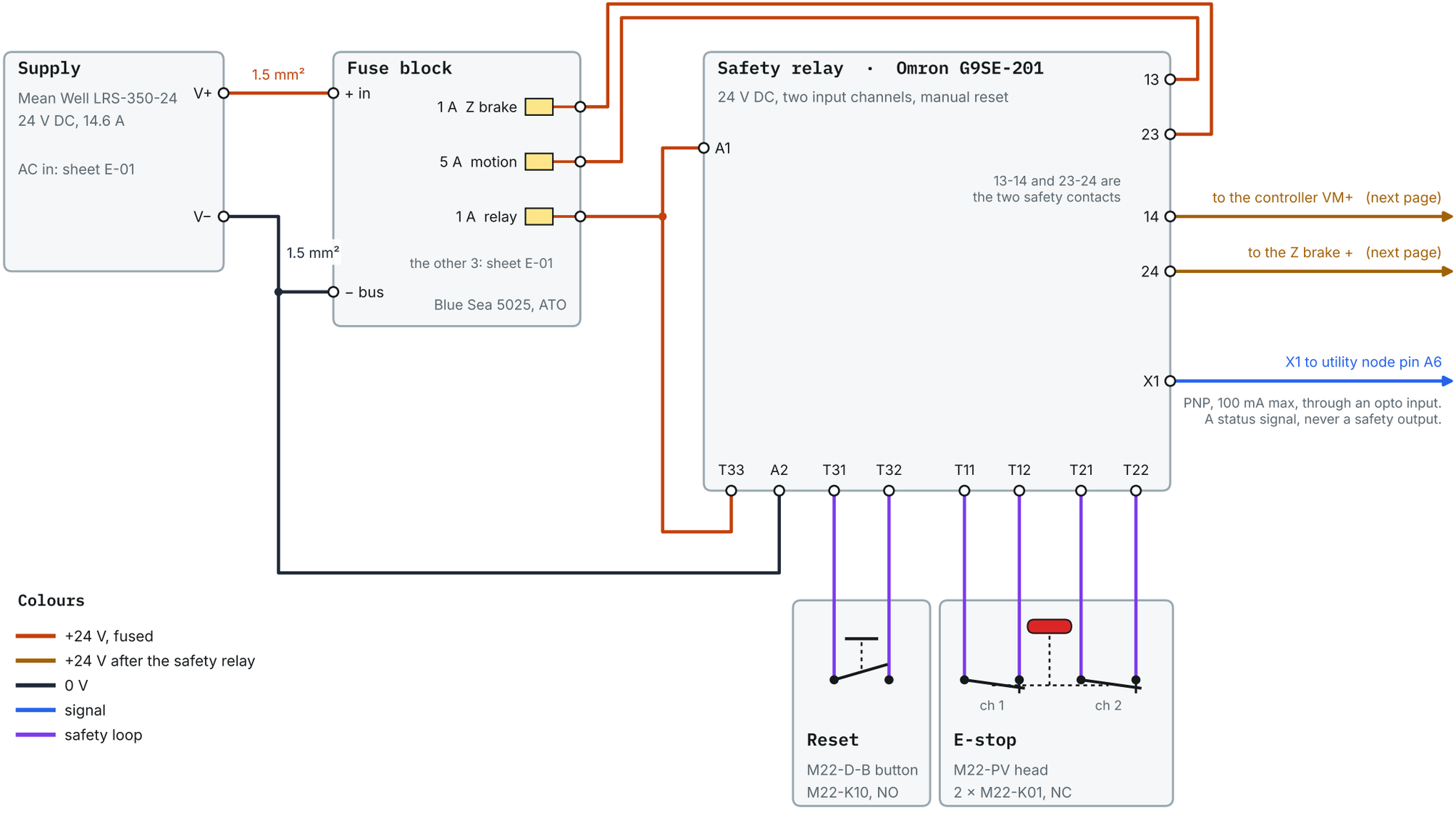
- 4× M3 × 6 socket cap
- 1× Controller (Elecrow 4-axis FluidNC)
- 4× M3 hex standoff, male-female, 10
- 1× Mounting panel, inside the enclosure
- 4× M3 nut

2.5 mm hex · 5.5 mm

Standoffs through the panel, nuts underneath, board on top, screws down into the standoffs. Use the board's own four holes: the pattern drawn here is the footprint block, not the board's drawing. Never screw a board straight to the extrusion: its ground plane on the frame makes a loop with the motor shields. Connections: the wiring pages, next.

Wiring · power and the safety relay

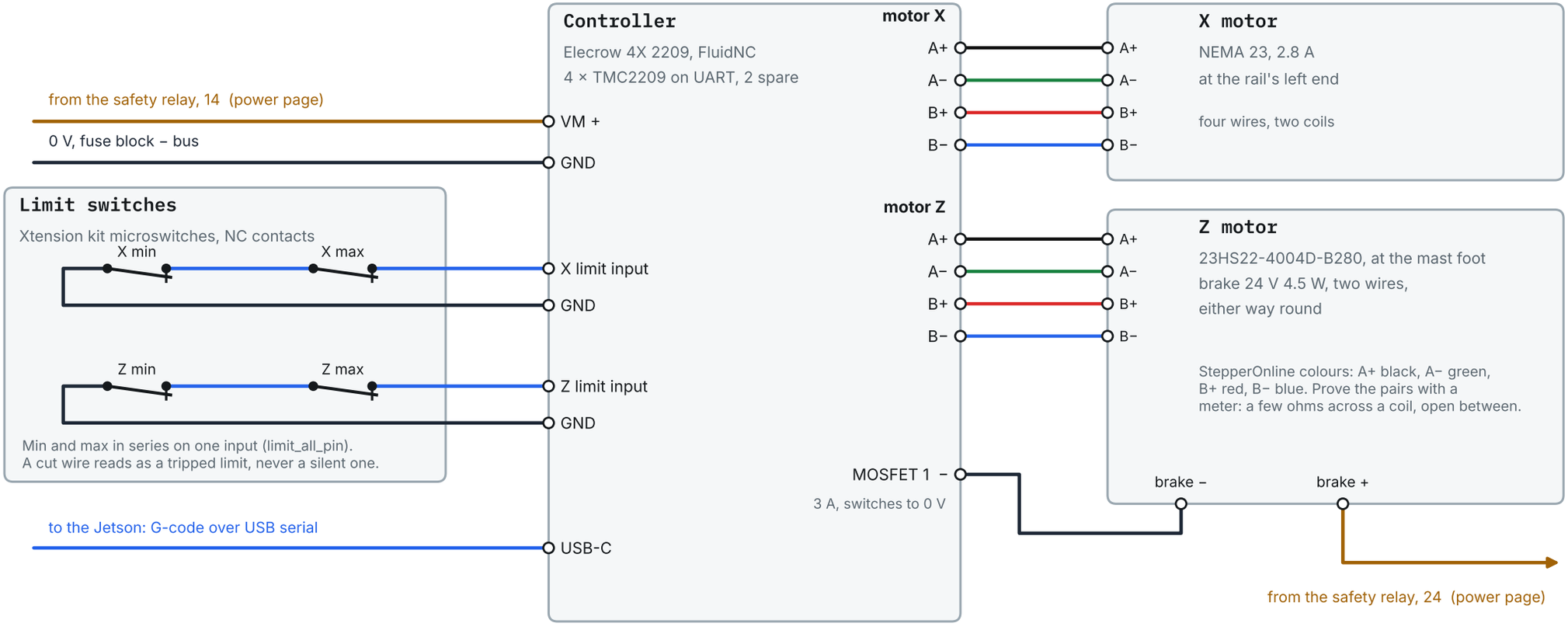
Gantry circuits only. Mains, the GFCI and the fused inlet are on sheet E-01.



How it resets	24 V on T33, the NO button from T31 to T32. After an E-stop the outputs stay off until the head is pulled up and the button is pressed and released.
Contact ratings	5 A at 30 V DC resistive, 1.5 A inductive. Omron limits the fuse ahead of each contact to 8 A: these are 5 A and 1 A.
Why two contacts	13-14 cuts the controller, so the motors go limp; 23-24 cuts the brake coil, so the mast locks.

Wiring · controller, motors, brake, limits

Elecrow 4X 2209 running FluidNC. Terminal names follow the board; GPIO numbers come from its config.



Driver settings (FluidNC config)

	run A	hold A	µsteps	steps/mm	drive
X	1.5	0.75	16	80	GT2 belt, 20 T pulley
Z	1.8	0.9	16	400	Tr8×8 lead screw

A TMC2209 is good for about 2 A RMS, so both motors run below rating.
Z lifts about 1.7 kg on an 8 mm lead: under 0.1 N·m.

What rides where

Cable	Path	Spec
X motor, X limits	along the rail, fixed	4 × 0.5 mm² shielded; 2 × 0.25
Z motor, brake, Z limits	X chain, then the mast	chainflex 4 × 0.5 shielded; 2 × 0.5; 2 × 0.25
Arm power, Cat6A, D405	X chain, then Z chain	arm brick lead; slim Cat6A; high-flex USB-C
VM, 0 V	enclosure only	1.5 mm², ferrules

Brake 24 V releases it. Before a Z move: M62 P0 (MOSFET on), then a 0.1 s dwell; after the move: M63 P0.
An E-stop or a power cut opens 23-24 and the mast locks whatever the software is doing.

Shields Motor cable shields to GND at the controller end only. No mains in any chain.

17 First motion, before the arm

Jog, calibrate steps, home. Nothing heavy on the gantry yet.

1. **Set driver current**
   X 1.5 A, Z 1.8 A run, half that to hold (wiring page). Too high cooks the TMC2209s.

2. **Jog 5 mm**
   confirm direction on both axes before anything longer.

3. **Command 100 mm**
   start at X 80, Z 400 steps/mm (16 µsteps), measure, correct until commanded equals actual.

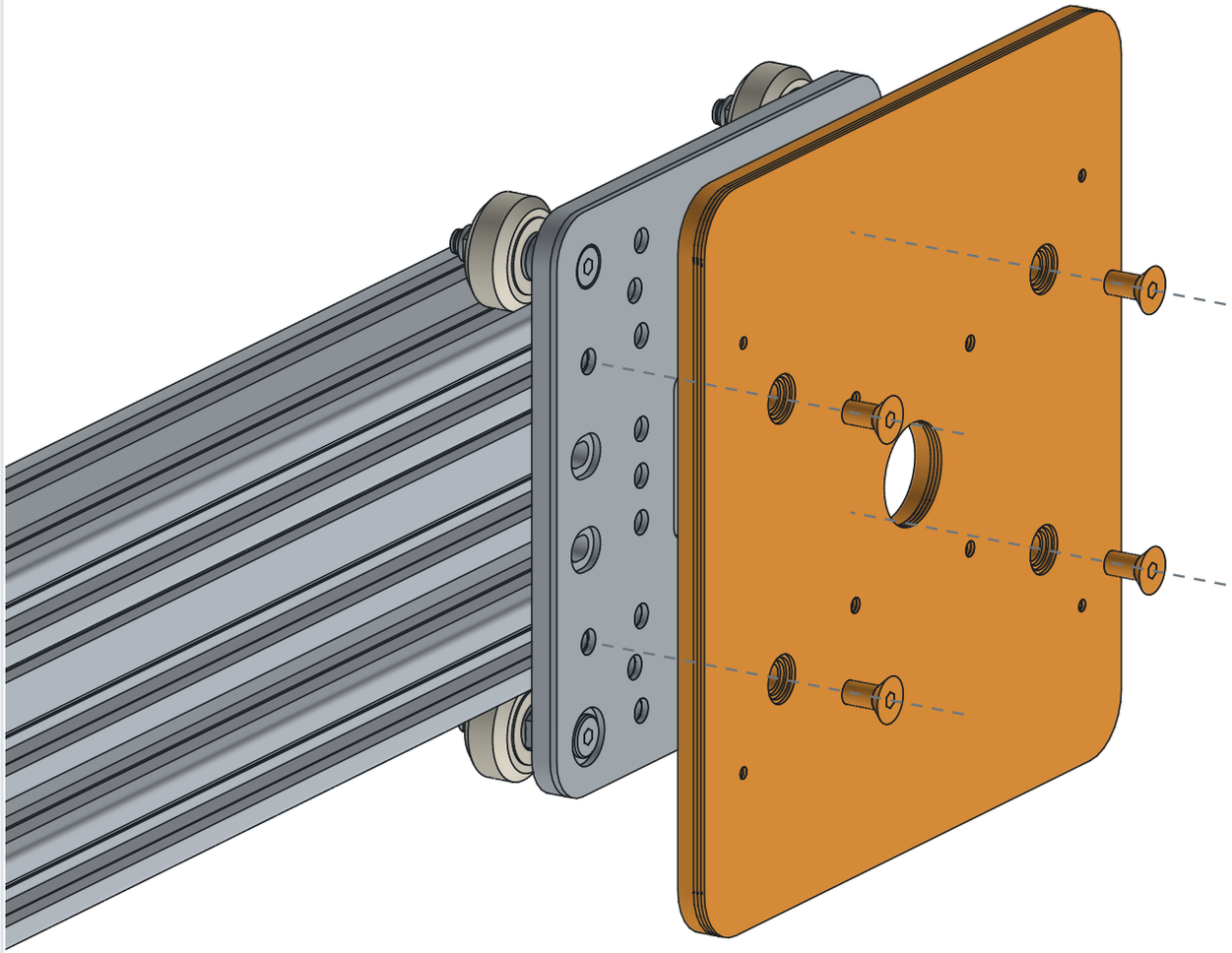
4. **Home both axes**
   and home on every power-up. FluidNC does not know where it is otherwise.

5. **Run full travel**
   ten times, both axes. Listen for a tight spot.

If the gantry is going to bind, lose steps or trip a limit wrongly, it does it now, when nothing expensive is bolted to it. That is the point of doing this before step 18.

18 Arm plate onto the Z carriage

Four M5 × 10 flat head into the plate's own tapped holes. Flush, so the arm sits flat.



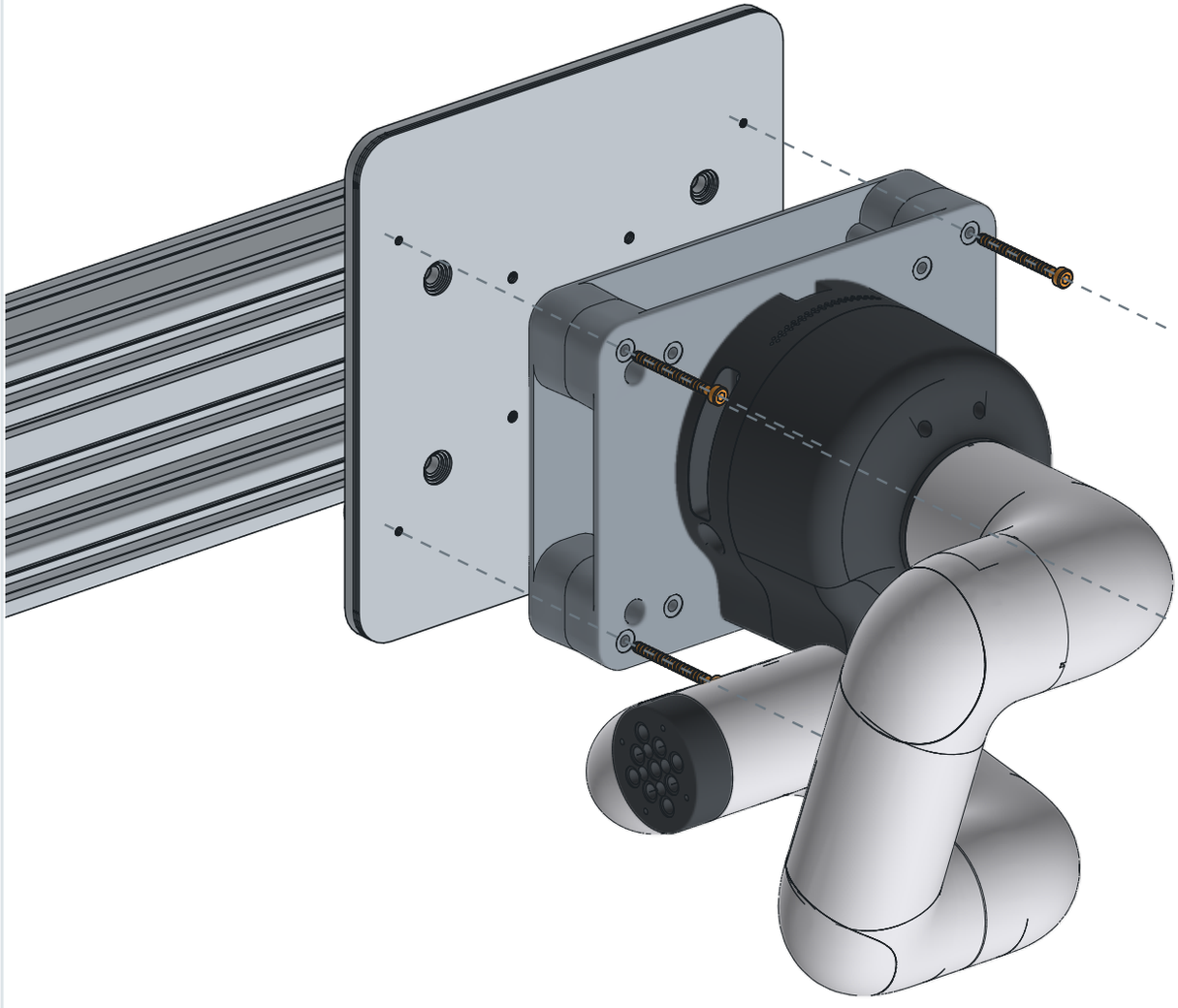
- 1× Arm plate Rev B, 160 × 160 × 6.35
- 4× M5 × 10 flat head, ISO 10642

3 mm hex

The XL plate has four tapped M5 holes at its corners, 100.6 apart along the mast and 60 across: the arm plate's countersunk holes land on them. Rev A's edge slots met no hole in the real plate. Order the Rev B file (arm_plate_revB.dxf) with the M3 and M4 holes TAPPED and the four M5 holes COUNTERSUNK 90° on the arm side: SendCutSend's drawing sizes are already in the file.

19 The arm, last

Its G base onto the plate: four M3 × 35 through the base's outer corner holes.



1× myCobot 280 Pi on its G base

4× M3 × 35 socket cap

SIDE mounted: J1 horizontal

2.5 mm hex, long

The G base's four outer holes are Ø 3.5 straight through its 32 mm; the arm plate has M3 taps under them. Unpowered, J1 turns by hand: swing the folded arm clear of each corner in turn. Without the G base, the arm bolts to the four M4 taps on the 44 mm square and sits 30 mm closer to the plate. Power it from its OWN brick, never the 24 V bus, and confirm it answers over TCP before any coordinated motion. Folded for every Z move: J2 −110, J3 −80, J4 −140, J5 −90.

20 Before you trust it

The safety circuit, and the checks that catch the usual faults.

- [ ] **E-stop wired, both channels**
  Two M22-K01 blocks into the G9SE-201. It cross-monitors them; one is not enough.

- [ ] **Two contacts, two jobs**
  13-14 feeds the controller, 23-24 the Z brake (wiring pages). Sensing and lighting stay alive.

- [ ] **E-stop mid-move**
  Motion stops, brake grips, no restart until the head is pulled up AND reset is pressed.

- [ ] **Power-cut test**
  Pull the plug mid-travel. The mast must hold, not fall.

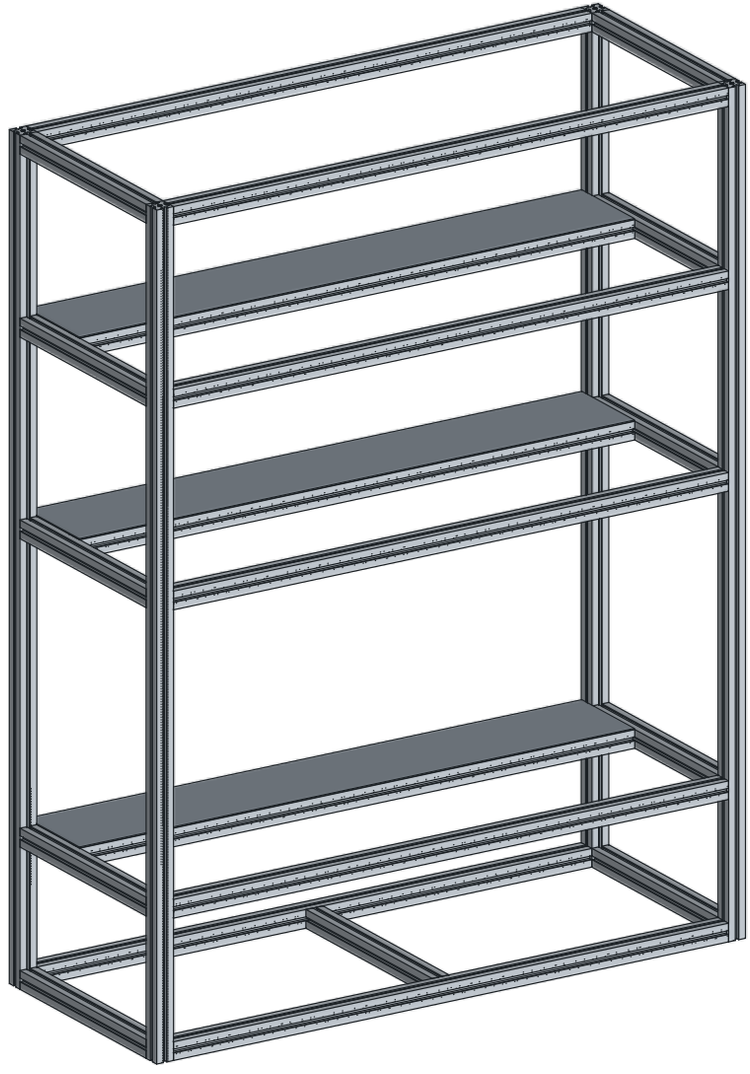
- [ ] **Limit test**
  Trip each switch by hand with the axis moving toward it.

- [ ] **Backlash check**
  Jog Z up, then down, to the same commanded point. Measure the difference.

Bench-testing the gantry without the safety relay is fine. Running the ARM without it is not.

21 The cabinet the gantry lands on

1397 × 559 × 1829, 40-series V-slot, five rail levels.



4 × 1829
uprights

13 × 1317
width rails: 5 front, 5 back, 3 under the tray's rear edge

11 × 479
depth rails (were 377 at 457 deep)

3 × 123
short rails under the tray laps

Tray line
y = 203 from the front. NOTHING of the frame behind it.

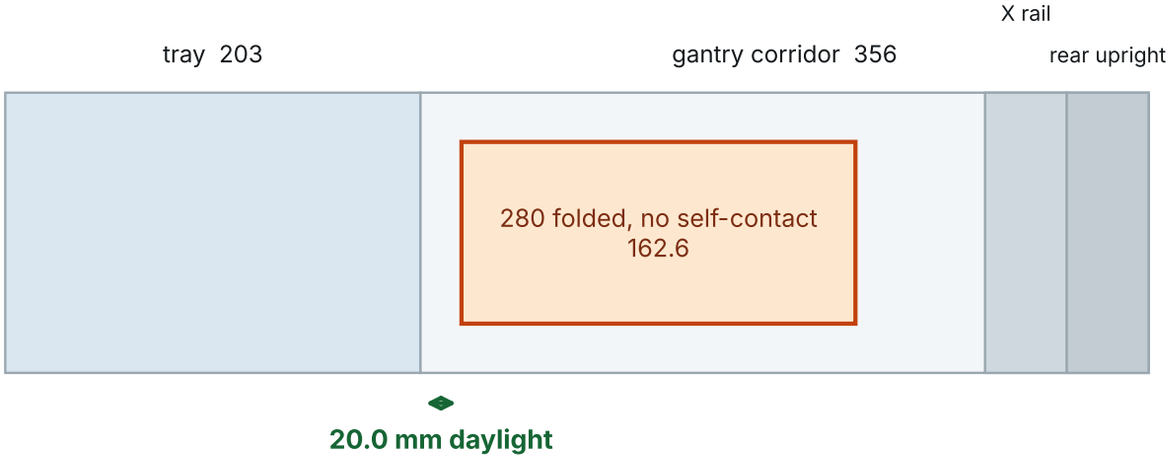
Corridor
356 mm, y = 203 to 559, all of it the gantry's

30.07 m of 4040, still 17 sticks of 2000 mm: 11 × [1317 + 479], 3 × [1829 + 123], 2 × [1317], 1 × [1829]. Only the eleven depth rails changed. The growing trays did not.

22 Why the cabinet got deeper

The one clearance that cannot be shimmed out later.

PLAN SECTION · front of the cabinet at the left · mm



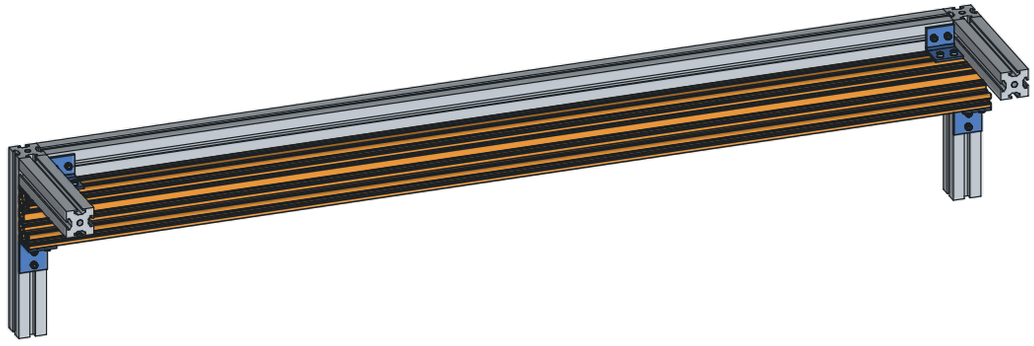
Cabinet depth	457 (Rev A)	533	559 (Rev E)
Arm J1 axis, from the front	283.6	359.6	385.6
Folded arm clears the tray by	−82.0	−6.0	20.0
Flange reaches forward to	−9.0	67.0	93.0
Gripper reaches forward to	−99.0	−23.0	3.0

The arm crosses a tray plane every time it changes tier, and the shortest fold it can reach without one link touching another is 162.6 mm, set by its own shoulder. At 457 deep it would overlap the tray by 82 mm. Measured on Elephant Robotics' mesh, collision checked.

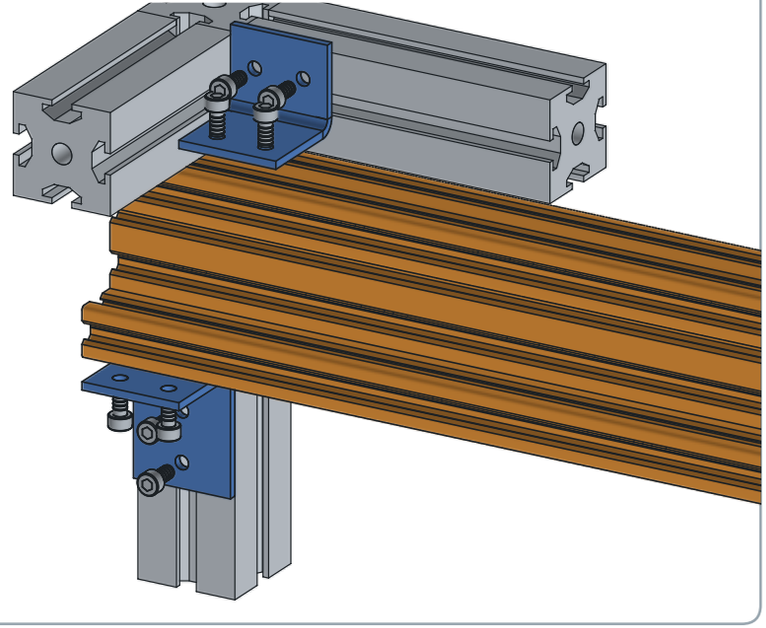
23 X rail onto the rear uprights

Cut to 1397. Two bent brackets at each end. This one must be level.

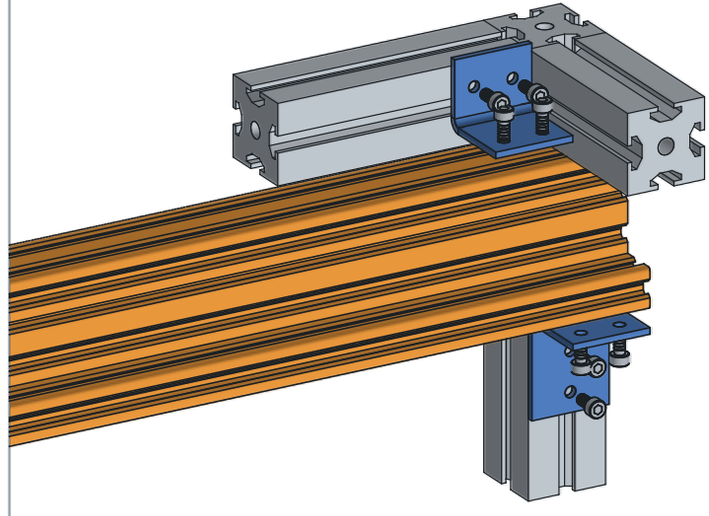
from the front, above



LEFT end, from the front-right: top bracket just inboard of the depth rail



RIGHT end, from the front-left: the same, mirrored



The rail's last 40 mm at each end sit UNDER the top depth rails, so nothing can bolt to its top face there. TOP bracket: leg on the rail's top face just inboard, 2 × M5 into T-nuts in the front top groove; leg up the top back rail's front face, 2 × M5 into slot-8 T-nuts. BOTTOM bracket: under the rail, 2 × M5 in the front bottom groove; leg down the rear upright, 2 × M5 into its slot. All M5 × 10. Level the top face before the last screw: tilt becomes height error across 1086 mm of travel.

24 Hang the Z mast

X carriage and X drive on the rail first, then the mast onto the carriage.

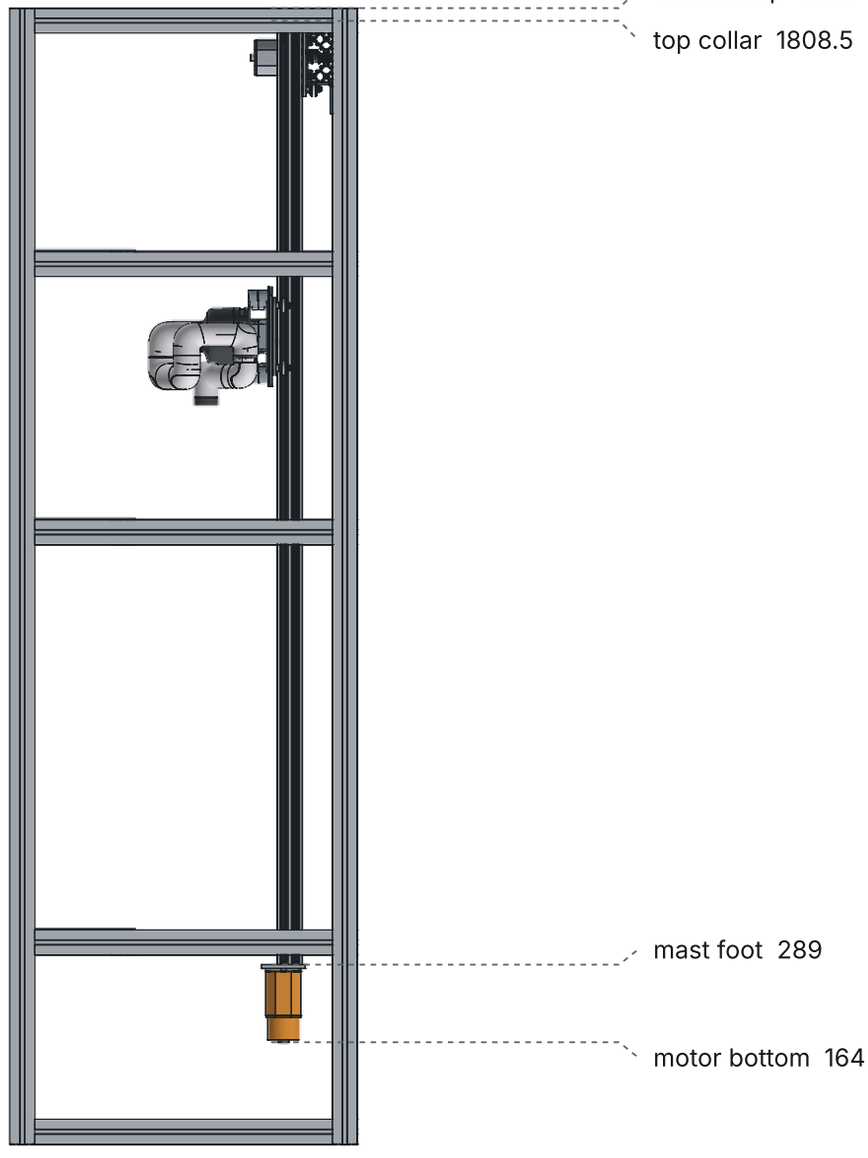


- 1× C-Beam mast, 1500, Z drive built
- 6× M5 × 10 low profile + T-nut

Six bolts through the X plate's holes at (±30, −30 / 0 / +30), heads on the rail side: low profile heads are 1.5 mm and the gap to the rail is 2.5. T-nuts in the mast's two outer back grooves. Top at 1789, foot at 289. Check plumb both ways with the carriage mid-rail, then at each end. If plumb changes across the travel, the rail is not level.

25 The Z drive, installed

Motor at the foot, screw hung from the top. Nothing above the cabinet.

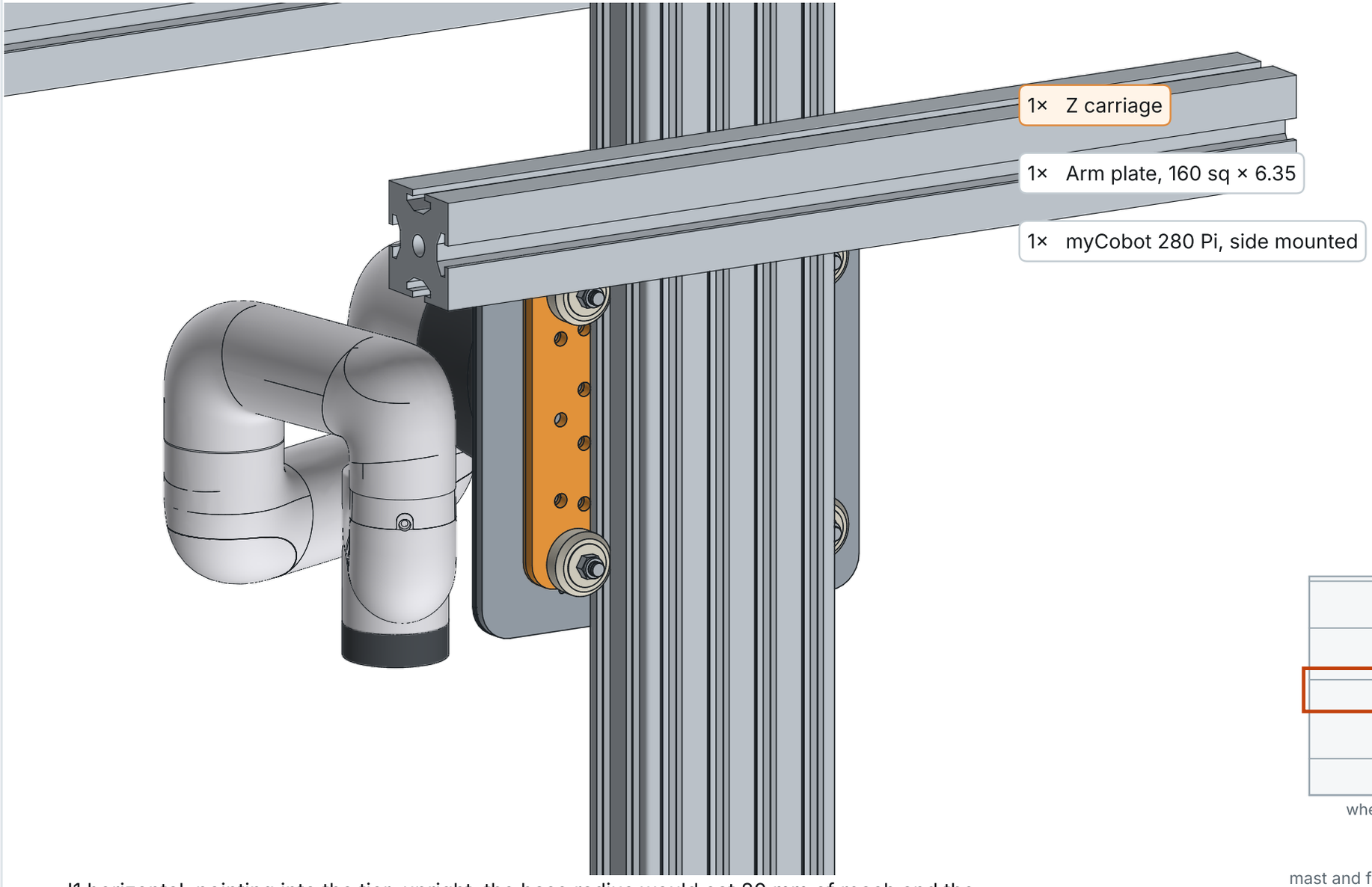


1× Braked NEMA 23 + coupler

Seen from the right-hand side, front of the cabinet to the LEFT. The top collar finishes 20.5 mm under the cabinet's top: a motor up there would stand proud of it. The screw hangs from the top bearings, so the coupler at the foot only ever carries torque.

26 Z carriage, arm plate, arm

Fold the arm before the first Z move, and keep it folded between tiers.





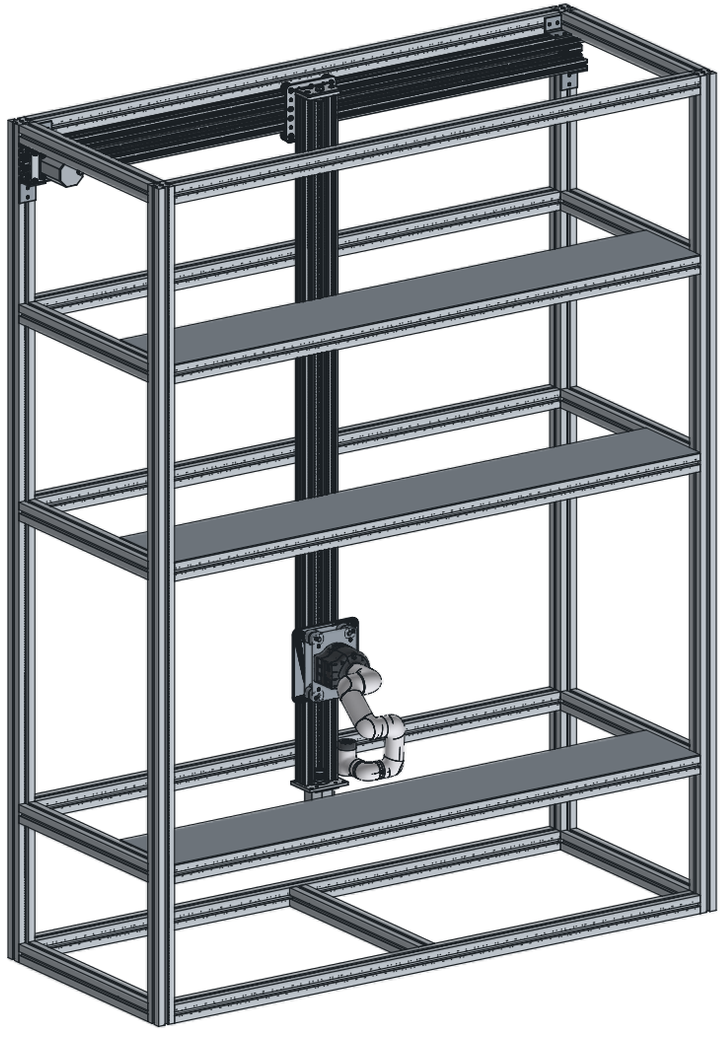
where this is

mast and frame shown cut

J1 horizontal, pointing into the tier: upright, the base radius would eat 80 mm of reach and the arm would stand 437 mm tall in a 341 mm opening. Fold: J2 −110, J3 −80, J4 −140, J5 −90, collision checked. An extended arm crossing a tray plane is the one crash this machine can do to itself at full torque.

27 Commissioned

Working envelope, and the six checks before you walk away.



X travel
1086, motor plate to idler

Z travel
1350 usable; tiers at 345, 1005, 1437

Flange reach
292.6 from J1, forward to y = 93

With the gripper
forward to y = 3: the whole tray

Folded transit
162.6 long, 20.0 clear of the tray line

Working height
263, inside the 341 tier opening

- [ ] Home X and Z, power-cycle, home again. Same zero both times.
- [ ] Fold the arm, run Z full travel. Nothing touches anything.
- [ ] E-stop mid-move: motion stops, brake grips, no reset until released.
- [ ] Pull the plug mid-travel. The mast holds.
- [ ] Jog to each tier, check the arm reaches the back row of pots.
- [ ] Hand-eye calibrate AT the tier it works at, not once at the top.

The mast is a 1.5 m cantilever off four V wheels. Expect 1 to 2 mm of forward lean at the foot that is not there at the top, which is why calibration is per tier.